

Day 6 — Saṁkrānti Moments

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify makara, karka, and other saṁkrāntis on the calendar.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sāyana* (IAST: sāyana; lit. “tropical zodiac (linked to seasons)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define saṁkrānti.** The moment the Sun moves from one rāśi to the next. ~12 saṁkrāntis per year (one every ~30 days). Two are most culturally prominent: *Makara Saṁkrānti* (Capricorn entry) ~Jan 14, and *Karka Saṁkrānti* (Cancer entry) ~Jul 16.
2. **The Jan 14 / Solstice mystery.** *Makara Saṁkrānti* is celebrated as the start of *Uttarāyaṇa* — but Dec 21 (solstice) is the astronomical start. Why the ~24-day difference? Answer (preview): the celebration uses the SIDEREAL (nirayana) zodiac, while the seasons (and solstice) align with the TROPICAL (sāyana) zodiac. Tomorrow we’ll unpack this.
3. **Source.** Sūrya-siddhānta computes saṁkrāntis as solar longitudes crossing rāśi boundaries.
4. **Cultural saṁkrāntis.** Festivals tied to specific saṁkrāntis: *Makara Saṁkrānti* (Pongal in Tamil Nadu, Lohri in Punjab, Magh Bihu in Assam), *Karka Saṁkrānti* (mid-monsoon).
5. **Class exercise.** Identify the 12 saṁkrānti dates for the current year (use a published pañcāṅga). Plot them on a circular year-wheel.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Saṃkrānti Moments.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).